

Algorithm: ATM Banking System

1. Start Program

1. Start execution of the program.
 2. Import required Java packages (`java.util.*`).
 3. Create a `Scanner` object for user input.
 4. Create a `HashMap<String, BankAccount>` to store bank users.
 5. Add predefined bank accounts with account number, PIN, and balance.
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2. Bank Account Setup

6. Create a `BankAccount` class with:
 - Account Number (`accNo`)
 - PIN (`pin`)
 - Balance (`balance`)
 - Transaction History (`ArrayList`)
 7. Initialize account using constructor.
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3. Display Main Menu

8. Display the main ATM menu:
 - 1. Login
 - 2. Exit
 9. Read user choice.
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4. Exit Condition

10. If user selects Exit:
 - Display “Thank You For Using ATM”
 - Terminate program.
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5. Login Process

11. If user selects Login:

- Input account number
- Input PIN

12. Validate credentials:

- Search account in `HashMap`
 - Call `login()` method
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6. Authentication Check

13. IF account is NOT found OR PIN is incorrect:

- Display “Login Failed”
- Return to main menu

14. ELSE:

- Display “Login Successful”
 - Proceed to ATM operations menu
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7. Enter ATM Menu Loop

15. Start loop until user logs out

16. Display ATM options:

- Check Balance
- Deposit
- Withdraw
- Transfer
- Transaction History
- Change PIN
- Logout